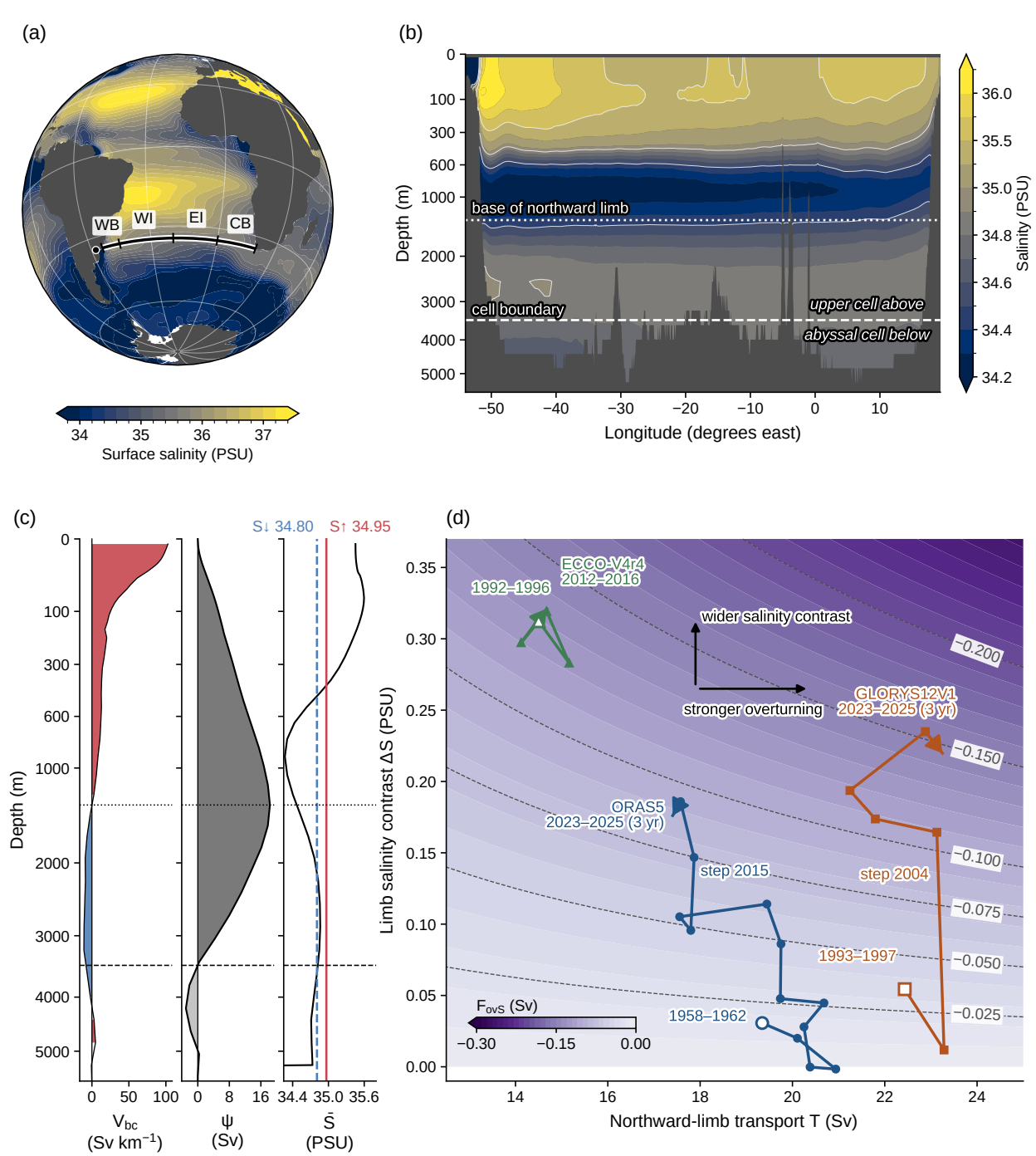

(a)
WB
WI
EI
CB
34
35
36
37
Surface salinity (PSU)
(b)
Depth (m)
base of northward limb
cell boundary
upper cell above
abyssal cell below
Longitude (degrees east)
Salinity (PSU)
(c)
S↓ 34.80
S↑ 34.95
Depth (m)
V_{bc} (Sv km^{-1})
ψ (Sv)
$\bar{S}$ (PSU)
(d)
ECCO-V4r4 2012–2016
1992–1996
wider salinity contrast
stronger overturning
GLORYS12V1 2023–2025 (3 yr)
ORAS5 2023–2025 (3 yr)
step 2015
step 2004
1993–1997
1958–1962
F_{ovS} (Sv)
−0.30
−0.15
0.00
−0.200
−0.150
−0.100
−0.075
−0.050
−0.025
Limb salinity contrast ΔS (PSU)
Northward-limb transport T (Sv)